

Georgios Kekkou

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 Giorgos Kekkou |  giorgoskekkou

Larnaca, Cyprus

RESEARCH INTERESTS

- Computer Architecture
- EDA: Testability & Reliability using ML

EDUCATION

- **University of Cyprus** September 2021 - May 2025
BSc, Computer Engineering Nicosia, Cyprus
 - GPA: 9.33/10.00 (Excellent)
 - Concentration area: Computer Hardware Systems and Intelligent Systems.
 - Technical Electives: Computer Architecture, CAD for VLSI, Digital Design with FPGA, Fault Tolerant Systems, Intelligent Systems, Digital Signal Processing.
- **Makarios III Lyceum of Larnaca** September 2017 - June 2020
Secondary Education Larnaca, Cyprus
 - Grade: 19.33/20 (Excellent)
 - Positive Sciences: Mathematics, Physics, Information Technology, Cisco Networks.

EXPERIENCE

- **Research Engineer** July 2025 - Present
KIOS Research and Innovation Center of Excellence  Nicosia, Cyprus
 - Review and analyze literature on VLSI testing problems to identify relevant research directions.
 - Explore potential innovative approaches that utilize modern machine learning techniques.
 - Reproduce and validate experimental results from prior studies to establish potential gaps for further research.
 - Research Area: VLSI Testing, Machine Learning.
- **Teaching Assistant** September 2025 - December 2025
Department of Electrical and Computer Engineering, University of Cyprus  Nicosia, Cyprus
 - Course in charge: Computer Aided Design (CAD) for VLSI.
 - Responsible for the lab sessions of the course.
 - Prepare and update lab materials including tutorials and exercises.
 - Review and mark student's assignments.
- **Summer Research Intern** June 2024 - July 2024
Texas A&M University  College Station, Texas, United States
 - Designed a converter tool that combines multiple serverless functions together.
 - Deployed Knative serverless functions to a Kubernetes-based cluster.
 - Apply compiler parsing ideas to design my tool.
 - Study research papers in the serverless computing domain.
- **QA Automation Engineer Intern** June 2023 - August 2023
HF Markets  Larnaca, Cyprus
 - Developed automated tests using the Selenium WebDriver to test web applications.
 - Designed customized pipelines for integration and testing using the Groovy programming language.
 - Executed and managed pipelines through the Jenkins CI/CD platform.
 - Developed automated tests for the mobile application using Selenium and Appium dedicated for Android and iOS.
 - API testing using Postman, plus integrating them in CI/CD pipelines using Newman.

- Additional: containerization using Docker, remote testing using Selenium Grid and Zalenium, automated alerts using Python, SQL database manipulation.

- **Passenger Handling**

Goldair Handling

Part-time summer job check-in passengers in the airport.

July 2022 - August 2022

Larnaca, Cyprus

- **Military Service**

National Guard of Cyprus

Mandatory military service for all Cypriot men after school.

July 2020 - August 2021

Cyprus

PROJECTS

- **Bachelor Thesis: Predicting hard-to-detect faults using Machine Learning**

September 2024 - May 2025

Tools: C++, Python, Multiprocessing, Machine Learning , VLSI Testing



- Developed framework for parsing digital circuit, converting to graph, and extracting structural features
- Implemented Parallel Fault Simulation that schedules the workflow on a multicore machine
- Constructed Graph Convolutional Network and Support Vector Machine models for classification
- Experimented with the developed Machine Learning models to predict hard-to-detect faults

- **Texas A&M Summer Internship Project: Serverless Workflow Function Converter**

June 2024 - July 2024

Tools: Python, Kubernetes (K3s), Docker, Virtual Machine



- Developed a compiler-based converter to convert serverless workflows
- Set up Knative cluster inside a VM to deploy serverless functions using K3s

- **Computer Architecture Lab 5: Cache Simulator**

November 2023 - December 2023

Tools: C++, Object-Oriented Programming (OOP)

- Cache Specifications: Single Word Blocks, Variable Number of Blocks and Ways
- Cache Organization: Direct Mapped, Set Associative, Fully Associative
- Cache Replacement: Least Recently Used (LRU), Static Re-Reference Interval Prediction (SRRIP)

- **Computer Architecture Labs 1-4: Processor Simulator**

September 2023 - December 2023

Tools: C++, Object-Oriented Programming (OOP), Parser, Implemented: ALU, Memory, Control Unit

Microarchitectures:

- Lab 1: Single Cycle
- Lab 2: Multi Cycle
- Lab 3: Pipeline - Hazard Free
- Lab 4: Pipeline - including Hazard Detection and Forwarding Units

SKILLS

- **Programming Languages:** C/C++, Python, Java
- **Programming Concepts:** OOP, Multiprocessing, Multithreading
- **Web Technologies:** HTML, CSS, JavaScript, Bootstrap, Flask
- **Database Systems:** MySQL, SQLite
- **Machine Learning:** PyTorch, Scikit-learn
- **Developer Tools:** Linux, Bash Scripting, Git, Makefile, Docker
- **Other Tools & Technologies:** Selenium WebDriver, CI/CD Pipelines

HONORS AND AWARDS

- Award for Best Academic Performance in Computer Engineering** June 2025
Faculty of Engineering at the University of Cyprus
- Award for Best Bachelor Thesis project in Computer Engineering** June 2025
Faculty of Engineering at the University of Cyprus
- Highest GPA among Computer Engineering Students** 2021-2025
Faculty of Engineering at the University of Cyprus
- Second Place in The 3-Minute CSR Video Challenge** April 2025
MENA Business Schools Alliance for Sustainability (MEBAS)
- Seeds for the Future Scholarship** March 2025
Huawei Technologies (Cyprus)
- Full Travel Grant Recipient** June 2024
uArch/ISCA 2024
- Scholarship for Higher Education studies** 2021-2025
Panagiotis and Ellis Malli Foundation
- Scholarship for Undergraduate Studies (IKYK)** 2021-2025
Cyprus State Scholarship Foundation
- Achieved 1st place in Cyprus-wide Pancyprrian Examinations (Computer Engineering, UCY)** July 2020
Pancyprrian Examinations 2020

PROFESSIONAL MEMBERSHIPS

- IEEE - Institute of Electrical and Electronics Engineers** August 2024 - Present
- IET - Institution of Engineering and Technology** October 2022 - Present

CERTIFICATIONS

- CS50x Introduction to Computer Science:** [View Certification](#) December 2022
- CCNA R&S: Routing and Switching Essentials** May 2020
- CCNA R&S: Introduction to Networks** May 2019

ADDITIONAL INFORMATION

Hobbies & Interests: Exploring computer hardware, learning about emerging technologies and fitness training.
Languages: English (proficient), Greek (native)